Ruchi Parekar

parekarruchi@gmail.com
GitHub
LinkedIn

+91-9371229702
Pimpri-Chinchwad, Pune, India

Summary

Electronics and Telecommunication Engineering student with practical experience in IoT, embedded systems, and PCB design. Proficient in Proteus simulation, microcontroller-based projects, and hardware debugging. Strong analytical and problem-solving skills with a keen interest in embedded and IoT technologies.

Education

- **JSPM's Rajarshi Shahu College of Engineering, Tathawade** — 2023-27
  Bachelor of Technology in Electronics and Telecommunication Engineering — CGPA: 9.21
- **SMT. S. S. Ajmera High School Junior College, Pune, India** — Passing: 2023
  Board – SSC, Science – Physics, Chemistry, and Mathematics — 75.50%
- **St. Alphonsa High School** — Passing: 2021
  Board – SSC — 85.40%

Experience

- **Project Intern- Sunay Enterprises, Pune** — July 2025

  This project was developed to address the challenges of water wastage and manual monitoring in domestic and industrial storage tanks. The system utilizes an ESP32 microcontroller and an ultrasonic sensor to continuously measure liquid levels and provide real-time monitoring. By enabling accurate level detection and timely alerts, the solution improves operational efficiency, enhances safety, supports automation, and significantly reduces the need for human intervention in water management processes.

Projects

- **ESP32-Based Wireless LED Control System** — May 2026- June 2026
  - Developed a wireless LED control system using ESP32 and an Android application.
  - Implemented Wi-Fi communication for remote LED control through a mobile device.
  - Achieved reliable real-time control, demonstrating IoT connectivity and embedded system integration.
- **IoT-Based Smart Liquid Level Monitoring System** — June 2025-July 2025
  - Identified the need to prevent liquid overflow, wastage, and manual monitoring in domestic and industrial tanks.
  - To design a smart IoT-based system for continuous liquid level monitoring with real-time alerts and automation.
  - Achieved accurate level detection and instant alerts, improving efficiency, safety, and reducing human intervention.name for this project.
- **PIC18F4550-Based Calculator System** — March 2025-May 2025
  - Developed an embedded calculator using PIC18F4550, keypad, and LCD display.
  - Implemented basic arithmetic operations: addition, subtraction, multiplication, and division.
  - Achieved accurate calculations with real-time result display in Proteus simulation.

Skills

- **Soft Skills:** Skilled in clear and persuasive communication, analytical problem-solving, cooperative teamwork, maintaining composure under pressure, delivering client-focused solutions, and ensuring precision in every task.
- **Technical Skills:** Intermediate C/C++, Python (Basic), Embedded C
- **Tools:** MATLAB, Arduino IDE, STM Cube IDE, Proteus, Simulink (Basic), MPLAB IDE

Position And Responsibility

Training and Placement Coordinator | E&TC Department
Student Advisor | IEI Student Chapter
Vice President | ENTESA

Certificate

Embedded Systems Bare-Metal Programming Ground Up (STM32)
STM32 with Embedded C Workshop